

Logarithmic Functions & Properties



Converting between exponential and logarithmic form

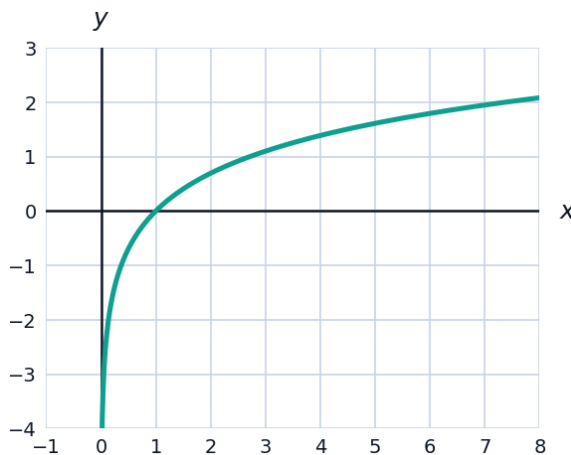
- 1 Rewrite $2^5 = 32$ in logarithmic form, and $\log_3 81 = 4$ in exponential form.
- 2 Evaluate $\log_5 125$ and $\log_2 \frac{1}{8}$ without a calculator.

Properties of logarithms

- 3 Use logarithm properties to expand $\log \left(\frac{x^3 \sqrt{y}}{z^2} \right)$ fully.
- 4 Combine $2\log x - \frac{1}{3}\log y + \log z$ into a single logarithm.
- 5 Use the change of base formula to evaluate $\log_7 50$, to three decimal places.

Visualizing logarithmic functions

- 6 The graph shows $f(x) = \ln(x)$, the natural logarithm.



- a State the domain and the vertical asymptote shown in the graph.
- b State the x-intercept shown.

Solving exponential equations using logarithms

- 7 Solve $5^x = 40$ for x , to three decimal places.

- 8 Solve $3(2)^{x-1} = 45$ for x , to three decimal places.
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Solving logarithmic equations

- 9 Solve $\log_2(x + 3) = 4$ for x , and check that the solution keeps the argument positive.
- 10 Solve $\log(x) + \log(x - 3) = 1$ for x , checking both solutions against the domain.
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Applications: the pH scale and decibels

- 11 The pH of a solution is given by $\text{pH} = -\log[H^+]$, where $[H^+]$ is the hydrogen ion concentration in mol/L. Find the pH of a solution with $[H^+] = 3.2 \times 10^{-5}$ mol/L, to two decimal places.
- 12 Sound intensity level in decibels is $L = 10\log\left(\frac{I}{I_0}\right)$, where I_0 is a reference intensity. If one sound is 1000 times more intense than another, find the difference in their decibel levels.
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Transformations of logarithmic functions

- 13 Describe how the graph of $g(x) = \ln(x + 2) - 3$ is transformed from the parent function $f(x) = \ln(x)$, and state its new vertical asymptote and domain.
- 14 Find the domain of $h(x) = \log(4 - x^2)$.
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Solving equations that combine exponentials and logarithms

- 15 Solve $e^{2x} - 5e^x + 6 = 0$ by substituting $u = e^x$.
- 16 Solve $\log_2(x) + \log_2(x - 2) = 3$ for x , checking the domain.
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Comparing logarithmic growth to other function families

- 17 Explain why logarithmic functions grow much more slowly than linear functions for large x , and give a numerical comparison of $\log(x)$ and x at $x = 1000$.
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Modeling with logarithms: the Richter scale

- 18** Earthquake magnitude on the Richter scale is $M = \log \left(\frac{I}{I_0} \right)$, where I_0 is a reference intensity. An earthquake measures magnitude 7, while another measures magnitude 5. Find how many times more intense the magnitude-7 earthquake is.
- 19** If an earthquake's intensity is 10,000 times a reference intensity I_0 , find its Richter magnitude.